

TANDIL, Argentina

WMO: 876450

Lat: 37.2411S

Lon: 59.2333W

Elev: 175

StdP: 99.24

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.2	-1.9	-7.0	2.1	3.8	-5.3	2.5	3.3	12.5	8.1	11.0	8.2	1.2	230	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	31.8	20.1	30.2	19.7	28.8	19.2	22.4	27.9	21.5	26.8	20.7	25.9	5.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	15.8	24.5	20.0	15.0	23.4	19.1	14.2	22.8	67.0	28.0	63.6	26.9	60.7	26.1	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.7	DB	-6.3	34.8	1.8	1.5	-7.6	35.8	-8.7	36.7	-9.7	37.5	-11.0	38.6	(4)
(5)			WB	-6.5	24.5	1.4	1.1	-7.5	25.3	-8.3	25.9	-9.1	26.5	-10.1	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	20.6	19.7	17.4	13.8	10.6	7.6	6.8	8.7	10.3	13.2	16.3	19.3	(6)	
	DBStd	5.73	3.12	3.27	3.38	3.33	3.34	2.97	3.31	3.48	3.19	3.22	3.30	3.41	(7)	
	HDD10.0	338	0	0	0	6	32	82	108	67	34	9	1	0	(8)	
	HDD18.3	1961	14	23	58	140	239	321	356	300	241	161	78	30	(9)	
	CDD10.0	1681	330	271	230	120	51	11	11	26	43	109	190	288	(10)	
	CDD18.3	261	86	61	30	4	1	0	0	1	1	2	17	60	(11)	
	CDH23.3	2872	963	577	258	42	4	0	0	4	6	39	233	747	(12)	
(13)	CDH26.7	883	340	175	52	4	0	0	0	1	1	4	49	257	(13)	
(14) Wind	WSAvg	3.8	3.9	3.6	3.4	3.2	3.1	3.4	3.6	4.0	4.3	4.3	4.1	4.2	(14)	
(15) Precipitation	PrecAvg	969	119	88	106	82	75	47	43	44	58	103	91	93	(15)	
	PrecMax	1264	216	207	301	442	215	139	137	154	170	202	220	176	(16)	
	PrecMin	587	39	19	26	0	3	0	1	0	10	40	21	15	(17)	
	PrecStd	173	56	44	66	93	61	36	36	36	42	45	53	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	33.0	30.3	26.8	23.5	18.3	19.0	22.8	23.6	26.8	30.2	33.7	(19)	
		MCWB	20.5	20.8	20.5	17.6	17.1	14.6	14.3	16.3	15.4	17.6	18.8	20.3	(20)	
	2%	DB	31.9	30.6	28.2	24.3	20.1	16.2	16.0	19.2	20.9	24.1	28.1	31.4	(21)	
		MCWB	20.5	20.5	19.6	17.4	15.1	12.4	12.0	13.5	13.6	16.4	17.8	19.2	(22)	
	5%	DB	30.2	28.9	26.2	22.4	18.0	14.6	14.3	16.9	19.0	22.2	26.1	29.6	(23)	
		MCWB	19.8	20.0	18.9	16.2	14.0	11.2	11.0	11.9	12.7	15.4	17.1	18.8	(24)	
	10%	DB	28.4	27.0	24.5	20.5	16.5	13.2	12.6	15.0	17.2	20.3	24.0	27.6	(25)	
		MCWB	19.1	19.3	18.2	15.7	13.4	10.5	10.0	10.8	11.7	14.4	16.3	18.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.9	23.7	22.6	21.1	18.7	16.8	16.2	18.2	18.1	19.7	20.5	22.8	(27)	
		MCDB	30.3	28.6	26.5	23.8	21.4	17.4	17.9	21.4	20.9	24.1	26.7	29.2	(28)	
	2%	WB	22.3	22.3	21.3	19.3	16.8	14.2	13.9	15.3	15.5	17.8	19.3	21.4	(29)	
		MCDB	28.5	27.0	25.4	21.5	18.4	15.0	14.7	17.3	18.7	22.0	25.4	27.8	(30)	
	5%	WB	21.3	21.3	20.2	17.8	15.4	12.3	11.9	13.3	14.0	16.5	18.2	20.3	(31)	
		MCDB	27.2	26.1	24.4	20.5	17.1	13.4	13.2	15.5	16.9	20.6	23.9	26.4	(32)	
	10%	WB	20.5	20.4	19.2	16.6	14.0	10.8	10.4	11.7	12.7	15.4	17.3	19.2	(33)	
		MCDB	25.9	25.0	23.1	19.6	15.7	12.5	12.0	14.4	15.8	18.7	22.3	25.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	12.8	12.1	11.8	10.6	10.2	10.3	11.0	11.7	11.7	13.0	14.0	(35)	
		MCDBR	16.4	15.4	14.3	14.7	12.6	11.9	12.2	14.1	15.3	15.1	16.0	16.8	(36)	
	5% WB	MCWBR	6.7	6.6	7.0	7.9	7.6	7.7	7.9	8.4	8.6	8.2	7.2	6.6	(37)	
		MCWBR	13.3	12.2	11.4	10.8	9.5	8.1	8.7	10.8	11.8	12.4	13.5	13.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.406	0.391	0.372	0.355	0.351	0.332	0.335	0.385	0.425	0.400	0.385	0.392	(40)	
		taud	2.298	2.353	2.396	2.401	2.370	2.435	2.405	2.233	2.116	2.245	2.317	2.317	(41)	
	Ebn at Noon		927	918	894	844	779	771	791	794	823	898	942	946	(42)	
		Edh at Noon	138	125	111	97	87	76	82	113	144	138	135	137	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.36	6.36	5.05	3.68	2.49	2.05	2.17	2.99	4.14	5.43	6.80	7.73	(44)	
	RadStd		0.35	0.45	0.32	0.37	0.21	0.20	0.21	0.33	0.31	0.55	0.37	0.37	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.75	N/A	N/A	N/A	N/A	N/A	+37
(47) Regional (3 neighbors)		N/A	N/A	N/A	+0.70	N/A	N/A	N/A	N/A	N/A	+33

Nomenclature: See separate page